

Mostafa Zaman

Researcher in AI, Control & Cyber-Physical Systems

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 Google Scholar
  Mostafa-Zaman
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  OpenCyberCity

Profile

Electrical and computer engineer developing hierarchical, constraint-aware, and uncertainty-informed learning and control methods for complex cyber-physical systems. Research applications span smart buildings, intelligent transportation, EV charging, microgrids, energy systems, IoT, and smart healthcare, with an emphasis on end-to-end experimental validation.

Research Interests

Methods: Hierarchical and safe reinforcement learning; bi-level and multi-objective optimization; model predictive control; probabilistic forecasting; anomaly detection

Systems: Smart cities; cyber-physical systems; smart buildings; intelligent transportation; smart grids and microgrids; IoT and sensor networks

Practice: Testbed design; sensing and embedded integration; data pipelines; closed-loop experimentation; reproducible research

Education

Ph.D. Virginia Commonwealth University , Electrical and Computer Engineering Dissertation — A Hierarchical Reinforcement Learning-Based Framework for Smart Cities Management • Advisor — Dr. Sherif Abdelwahed • Multi-level, multi-objective deep reinforcement learning for coordinated smart buildings, transportation, and healthcare	Richmond, Virginia, USA Dec 2022 – 2026
M.S. Virginia Commonwealth University , Electrical and Computer Engineering	Richmond, Virginia, USA Aug 2018 – Dec 2022
B.S. Bangladesh University of Engineering and Technology , Electrical and Electronic Engineering	Dhaka, Bangladesh May 2010 – Sept 2015

Research Experience

Virginia Commonwealth University , Graduate Research Assistant — OpenCyberCity Developed intelligent control, optimization, data analytics, and experimental infrastructure for a modular smart-city cyber-physical testbed. • Built learning and optimization pipelines for smart buildings, transportation, water, energy, and healthcare applications. • Developed bi-level, multi-objective, and reinforcement-learning components for inter-connected infrastructure under operational constraints. • Integrated sensors, embedded hardware, middleware, data pipelines, dashboards, and closed-loop control systems. • Mentored more than 20 high-school, undergraduate, and graduate researchers. • Contributed to peer-reviewed publications, posters, demonstrations, and grant-related activities.	Richmond, Virginia, USA Aug 2018 – 2026
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Professional Experience

Energypac Engineering Limited , Assistant Engineer (R&D), Power Transformer Division	Dhaka, Bangladesh
- Designed 132/11 kV and 33/11 kV power transformers, including general arrangements, guaranteed technical particulars, and nameplates. - Managed core-coil assembly requirements, conductor and core requisitions, OLTC specifications, and winding details. - Prepared costing sheets and technical documentation for transformer-manufacturing projects. - Completed ABB training on safety of on-load tap changers and transformer protection.	Mar 2016 – June 2018

Teaching Experience

Virginia Commonwealth University , Teaching Assistant, Electrical and Computer Engineering	Richmond, Virginia, USA
Supported instruction, laboratories, assessment, office hours, and student mentoring across core ECE courses.	Aug 2018 – 2026
- Signals and Systems — continuous/discrete-time systems, convolution, Fourier analysis, Laplace and Z transforms. - Microcomputer Systems — Vivado workflows, simulation, hardware interfaces, and real-time applications. - Digital Logic Design — five semesters of digital circuits and FPGA laboratories. - Engineering Programming — five semesters of C programming, algorithms, and structured problem solving. - Industrial Automation, Digital Systems, and Advanced Engineering Programming.	

Selected Publications

A Review of IoT-Based Smart City Development and Management	June 2024
M. Zaman, N. Puryear, S. Abdelwahed, N. Zohrabi 10.3390/smartcities7030061 (Smart Cities)	
A Multi-Level Reinforcement Learning Framework for Smart Building Management	Apr 2026
M. Zaman, S. Iannucci, N. Zohrabi, S. Abdelwahed (13th IEEE Conference on Technologies for Sustainability (SusTech))	
Graph-Based Bi-Level Reinforcement Learning for Coordinated EV Fast Charging Under Distribution Constraints	2026
M. Zaman, S. Abdelwahed (IEEE Transportation Electrification Conference & Expo (ITEC))	
Integrative ITS Platform for Enhanced Urban Mobility in Smart Cities	Aug 2025
M. Zaman, A. Malik, D. Gubay, A. DeWitt, S. Varzaneh, N. Zohrabi, S. Abdelwahed (IEEE International Conference on Ubiquitous Intelligence and Computing)	
A Machine Learning-Based Temperature Control and Security Protection for Smart Buildings	July 2024
M. Zaman, M. Al Islam, N. Zohrabi, S. Abdelwahed 10.1109/SMARTCOMP61445.2024.00070 (IEEE International Conference on Smart Computing (SMARTCOMP))	
Demand-Response Prediction in Smart Grids Using Machine Learning Techniques	Feb 2024
M. Zaman, S. Saha, N. Zohrabi, S. Abdelwahed 10.1109/ISGT59692.2024.10454224 (IEEE Power & Energy Society Innovative Smart Grid Technologies Conference)	

Optimizing Smart City Water Distribution Systems Using Deep Reinforcement Learning M. Zaman, A. Tantawy, S. Abdelwahed 10.1109/HONET59747.2023.10374894 (IEEE HONET)	Dec 2023
A Decision Support System for Cyber Physical Systems under Disruptive Events — Smart Building Application M. Zaman, R. Eini, N. Zohrabi, S. Abdelwahed 10.1109/ISC255366.2022.9922493 (IEEE International Smart Cities Conference)	Sept 2022
Uncertainty Estimation in Power Consumption of a Smart Home Using Bayesian LSTM Networks M. Zaman, S. Saha, N. Zohrabi, S. Abdelwahed 10.1109/AdCONIP55568.2022.9894187 (IEEE International Symposium on Advanced Control of Industrial Processes)	Aug 2022
Adaptive Control for Smart Water Distribution Systems M. Zaman, M. Al Islam, A. Tantawy, C. J. Fung, S. Abdelwahed 10.1109/ISC253183.2021.9562812 (IEEE International Smart Cities Conference)	Sept 2021

Technical Skills

Programming: Python; MATLAB; C; JavaScript; LaTeX

Machine Learning & AI: PyTorch; TensorFlow; Keras; Scikit-learn; deep reinforcement learning; Bayesian deep learning

Data & Visualization: Pandas; NumPy; Matplotlib; Seaborn; Jupyter

Control & Simulation: Simulink; model predictive control; PSpice; OpenMarkov; PSAT

Embedded & Digital Systems: Sensor integration; FPGA development; Vivado; VHDL; microcontroller programming; Proteus

Development: Git/GitHub; Linux; HPC clusters; Visual Studio; reproducible experimentation

Grants & Research Programs

CCI CVN Smart City Testbed — Phase 2 , Graduate Research Assistant • Supported expansion of OpenCyberCity services, transportation components, and data and analytics capabilities.	Richmond, Virginia, USA Aug 2022 – 2026
NSF Opioid-Crisis Planning Project , Graduate Research Assistant • Contributed sensing, data-collection, predictive, and explainable-analytics planning for community-informed decision support.	Richmond, Virginia, USA Dec 2022 – July 2023
VCU–Meru Biotech Collaborative Research , Graduate Research Assistant • Supported sensing-system development and experimental validation for molecular-binding detection.	Richmond, Virginia, USA Jan 2022 – Aug 2022

Honors & Awards

Outstanding Graduate Research Award — VCU College of Engineering Recognition for research excellence, leadership, mentorship, and contributions to OpenCyberCity.	2025
Best Poster Award — IEEE SMARTCOMP OpenCyberCity testbed progress in smart-city management.	June 2024
Travel Award — IEEE Transportation Electrification Conference & Expo	2022
Travel Awards — VCU ECE Department and College of Engineering	2021–2022

Dean's List — Undergraduate Program

Professional Service

Journal reviewer: IEEE Transactions on Artificial Intelligence; IEEE Open Journal of Control Systems; Frontiers in Energy Research — Smart Grids; Journal of Ambient Intelligence and Humanized Computing

Conference reviewer: IEEE CCTA; IEEE IGESSC; IEEE ISC2; IEEE HONET; IEEE CICON; HPCS; CSNT; CSAE

Memberships: IEEE; IEEE Young Professionals; IEEE Control Systems Society; IEEE Power Electronics Society; IEEE Systems Council

Mentoring

Student research mentoring

Mentored more than 20 high-school, undergraduate, and graduate researchers in problem formulation, algorithm development, implementation, experimental validation, writing, and presentation.